Journal of Applied Physics

RESEARCH ARTICLE | SEPTEMBER 18 2025

Design studies on large muon trigger detectors using silicon photomultiplier devices

Special Collection: Muography: Discoveries, Innovations, and Applications

G. Rao ; A. Bieberle ; U. Hampel ; M. Wagner

Check for updates

J. Appl. Phys. 138, 114504 (2025)
https://doi.org/10.1063/5.0288270

View Online
Export Citation

Articles You May Be Interested In

Neutron radiation hardness tests of SiPMs

AIP Conf. Proc. (October 2013)

Performance Evaluation of SiPM Photosensors in the Presence of Magnetic Fields

AIP Conf. Proc. (April 2010)

High-Resolution Muography Using a Prototype Portable Muon Telescope

J. Undergrad. Rep. Phys. (January 2020)

21 October 2025 08:11:13

Nanotechnology & Materials Science
Optics & Photonics
Impedance Analysis
Scanning Probe Microscopy
Sensors
Failure Analysis & Semiconductors



Unlock the Full Spectrum.
From DC to 8.5 GHz.
Your Application. Measured.
Find out more
Zurich Instruments

AIP Publishing

Journal of Applied Physics ARTICLE pubs.aip.org/aip/jap

Design studies on large muon trigger detectors using silicon photomultiplier devices

Cite as: J. Appl. Phys. **138**, 114504 (2025); doi: 10.1063/5.0288270
Submitted: 30 June 2025 · Accepted: 1 September 2025 ·
Published Online: 18 September 2025

View Online Export Citation CrossMark

G. Rao,[1,a)] A. Bieberle,[1,b)] U. Hampel,[2,c)] and M. Wagner[2]

AFFILIATIONS

[1]Institute of Fluid Dynamics, Helmholtz—Zentrum Dresden-Rossendorf, Bautzner Landstrasse 400, 01328 Dresden, Germany
[2]Chair of Imaging Techniques in Energy and Process Engineering, Dresden University of Technology, Dresden, Germany

Note: This paper is part of the Special Topic on Muography: Discoveries, Innovations, and Applications.
[a)]g.rao@hzdr.de
[b)]**Author to whom correspondence should be addressed:** a.bieberle@hzdr.de
[c)]**Also at:** Institute of Fluid Dynamics, Helmholtz—Zentrum Dresden-Rossendorf, Bautzner Landstrasse 400, 01328 Dresden, Germany.

ABSTRACT

This work presents the design and experimental evaluation of muon trigger detectors tailored for industrial-scale applications. Studies were conducted using an organic scintillator plate measuring $250 \times 250 \times 10\,\mathrm{mm}^3$, configured with varying numbers of silicon photomultiplier (SiPM) devices, each with a detection area of $4 \times 4\,\mathrm{mm}^2$. Various reflective materials, such as reflective paint, aluminum foil, and high-gloss paper, were applied to enhance the detected scintillation photon yield. Particular attention was given to optimize dark count suppression and muon event discrimination. Four detector configurations were designed, optimized, and evaluated individually. Spatially resolved muon detection efficiency was assessed using a muon coincidence detector with an active area of $50 \times 50\,\mathrm{mm}^2$. The assembly procedures for the different detector configurations are also discussed. Finally, a trigger detector configuration incorporating a single SiPM device coupled with a wavelength-shifting fiber clamp was identified as providing the most efficient collection for scintillation photon yield. This configuration is cost-effective, modular, and easy to mass-produce, making it well-suited for the developing of large-scale muon trigger detectors for industrial applications.

21 October 2025 08:11:13

© 2025 Author(s). All article content, except where otherwise noted, is licensed under a Creative Commons Attribution (CC BY) license (https://creativecommons.org/licenses/by/4.0/). https://doi.org/10.1063/5.0288270

I. INTRODUCTION

In recent decades, numerous muon detector systems have been developed for use in fundamental research and large-scale imaging applications. Notable examples in particle physics include ATLAS,[1] AMIGA,[2,3] PANDA,[4] and ALICE.[5] Muon imaging has also been employed to investigate large structures such as pyramids,[6–8] molten reactor core simulation,[9] volcanoes,[10] deep-ocean environments,[11] storage casks for high-level radioactive waste,[12–15] and for border security applications.[16] For imaging purposes, naturally occurring cosmic-ray muons are utilized. These muons are produced in the Earth's upper atmosphere through interactions between high-energy cosmic radiation and atmospheric nuclei.[17] Muons are secondary particles with a mass approximately 200 times that of electrons, and they travel at velocities close to the speed of light. Despite their relatively short lifetime of around $2.2\,\mu\mathrm{s}$, they reach the Earth's surface due to relativistic time dilation. At sea level, cosmic-ray muons have a mean energy of around 4 GeV and a flux of about $1\,\mathrm{muon\,cm^{-2}\,min^{-1}}$.[18] The flux distribution follows a zenith angle dependence proportional to $\cos^2\theta$.

Since cosmic-ray muons possess energies in the GeV range and have low radiation losses, they are well-suited for imaging large, high-density objects. Over the past five decades, numerous muon detector systems have been developed successfully for the detection of cosmic-ray muons and other high-energy particles. Consequently, the foundational knowledge of stand-alone muon detector technology for particle analysis is well-established.[19,20] However, most of these muon detection systems are designed as

© Author(s) 2025

fixed installations. They are typically heavy, bulky, and sensitive to mechanical disturbances. Therefore, their applicability as generic, modular detectors for object-specific applications is significantly limited.

The two primary technologies commonly employed for muon detection are ionization gas detectors and scintillation detectors.[19] Increasing the density of individual detectors over a detection area enables the spatial resolution of an incident muon to be improved to below 1 mm and an angular resolution of 5–10 mrad.[21] It is generally accepted that these resolution levels are sufficient for imaging large-scale structures. However, an increase in detector density inevitably leads to greater electronics complexity, as all the detector channels must be operated simultaneously. This makes the electronic units bulky, expensive, and energy-inefficient, restraining the construction of large-scale detectors for the measurement of low cosmic-ray muon flux.

A special ionization gas detector, known as drift chamber, offers large area coverage for detection. To determine the location of muon interactions, precise drift time t_{Drift} measurement is required. Therefore, suitable large trigger detectors are needed that provide a prompt start signal upon muon detection.[22–24] Large-area scintillation detectors are typically used for this purpose. Scintillation photons generated by muon interactions are promptly detected by photomultiplier tubes (PMTs), which are either surface-mounted or embedded within the scintillator slab. PMTs can amplify and convert single optical photons by a factor of approximately 10^7. Due to their nearly noise-free electron amplification, PMTs are well suited for detecting very low light levels.[19] A transimpedance amplifier subsequently converts the resulting charge into voltage pulses lasting $1-2\,\mu$s, with amplitudes proportional to the number of optical photons detected. However, PMTs require high supply voltages, are bulky and fragile, and are highly sensitive to mechanical vibrations. Consequently, their use is limited in industrial environments where access is restricted or detector mobility is required.

Over the last decade, silicon photomultipliers (SiPMs) have been developed to overcome many of the limitations associated with traditional PMTs. SiPMs are dense arrays of microscopic avalanche photodiodes, known as cells, which operate in Geiger mode[19] using a bias voltage typically between 28 and 40 V. SiPMs have already been used in large-scale muon detector systems, achieving spatial resolutions of up to 10 mm.[25–28] However, the active detection area of SiPMs is small on their exposed surface, limiting their coverage when they are coupled directly with a scintillator slab. To overcome this issue, they are often combined with wavelength-shifting (WLS) fibers. These fibers absorb scintillation photons from greater distances and emit their own photons, which have a slightly different wavelength. They then transport these photons directly onto the small active area of the SiPMs.

Replacing PMTs with SiPMs is not straightforward for large-area trigger detector applications. This is evident, for example, in the NeuLAND detector.[29] Although SiPMs can generate measurable signals from single-photon events, their high dark count rates make it difficult to distinguish signals under very low light conditions. Each SiPM cell is subject to random self-triggering caused by thermally generated electrons.[30] In such cases, the electrical signals generated by external photons and those caused by thermal noise are indistinguishable. Additionally, a triggered cell can induce secondary triggering in neighboring cells. Dark counts may also include minor contributions from environmental background radiation on the surface level. Thermal- and voltage-induced dark counts occur in the kilohertz range and dominate the noise spectrum. Therefore, to ensure reliable detection, a minimum number of optical photons must interact with the SiPM such that the resulting signal amplitude exceeds the thermal noise baseline. In the literature, a threshold of approximately 10 photons (or cell levels) is commonly recommended.[30]

This issue is particularly challenging when SiPM devices are used alongside large scintillator detectors for muon detection applications. The scintillation light reaching the SiPM surface from distant regions can be so weak that the resulting electrical signals fall within the range of self-triggered noise. Due to rates reaching several kHz/mm^2,[30] dark counts dominate the data acquisition process, significantly exceeding the contribution from the low flux of cosmic-ray muons. To increase the detected optical photon yield associated with each muon event, the detection area could be proportionally expanded; however, this also results in higher dark count rates. This, in turn, increases the likelihood of self-triggered pulses, meaning that higher voltage threshold settings are required to accurately distinguish true muon signals.

In this study, a scintillation plate is employed as a muon trigger detector to maximize photon acquisition from cosmic-ray muon hit events, coupled with SiPM devices in multiple configurations. These configurations were assembled and tested to validate their effectiveness in terms of high-efficiency muon detection. All investigations are based on an all-sides-polished organic scintillation plate measuring $250 \times 250 \times 10\,\mathrm{mm}^3$. Different large active areas are realized by coupling varying numbers of SiPMs to the plate's side faces. The SiPMs are connected electronically in parallel and share a single signal processing stage, which has been optimized accordingly. Furthermore, the impact of different reflective materials—optical reflector paint, commercially available aluminium foil, and high-gloss paper—on the detected optical photon yield for each detector configuration is investigated. To evaluate the scintillation plates at pre-determined regions of interest, a muon coincidence detector (MCD) with an active area of $50 \times 50\,\mathrm{mm}^2$, coupled with traditional photomultiplier tubes, is implemented in a modified design and repurposed to track muons specifically for the localized evaluation of the muon trigger detector. Finally, a general qualification method is proposed to evaluate SiPM-based muon trigger detectors under moderate pulse count rate conditions.

II. MATERIALS AND METHODS

A. Materials

Organic scintillators convert high-energy radiation into photons within a narrow optical wavelength range. These photons are then collected using photodetectors, which generate an electric charge proportional to the number of photons received. In pulse-counting mode, a charge-sensitive preamplifier delivers voltage pulses whose amplitude corresponds to the energy deposited. In order to achieve high detection efficiency, it is necessary for the maximum number of generated scintillation photons to interact

21 October 2025 08:11:13

© Author(s) 2025

with a solid-state photodetector. Therefore, the emission spectrum of the scintillation light must overlap significantly with the spectral sensitivity range of the photodetector. The primary objective in designing the muon trigger detector is to direct as many of the photons generated within the scintillator volume as possible onto the small active area of the solid-state photodetector. Designing large-area muon trigger detectors for industrial applications is particularly challenging, as the system must remain simple, modular, and flexible. This section presents the rationale behind the selection of components for the development of a large-area muon trigger detector. Furthermore, four configurations are described and evaluated with the aim of achieving a simple, cost-effective, and adaptable design.

The scintillation material and the dimensions of the trigger detector module were initially determined, as these are the parameters that have the greatest influence on manual handling. For this study, an organic scintillator plate UPS-92S (AdvTECH UK Limited®) with an active area of $250 \times 250\,\mathrm{mm}^2$ and a thickness of 10 mm was chosen. The key parameters of the scintillator material are summarized in Table I. The plate weighs 0.656 kg in total and yields approximately 8000–10 000 photons/MeV of deposited energy. Since muons with energy of 1–10 GeV roughly deplete 2 MeV/cm when passing through organic scintillator materials,[31] a detectable optical signal is expected across the entire surface from incident cosmic-ray muons. All sides of the plate were polished to enhance surface reflection and maximize scintillation light retention.

Silicon photomultipliers (SiPMs) from CEFLA S.C., with an active area of $4 \times 4\,\mathrm{mm}^2$, were selected for use as photodetectors (model ASD-NUV4S-P-40). The key parameters of the SiPMs are summarized in Table II. They provide a flat resin surface with a refractive index that closely matches that of the scintillation material, enabling optimal optical coupling. The SiPMs were operated with an overvoltage of 2.5 V, and all the experiments were conducted at a stable room temperature of 21 °C. The voltage was not corrected for any temperature drifts during the course of the experiment. Any residual surface roughness is compensated by applying optical grease (BC-630, BICRON Inc.) between the SiPM and the scintillator plate. The SiPM's spectral peak sensitivity closely aligns with the peak emission wavelength of the organic scintillator. Each SiPM contains 9340 individual cells, providing sufficient dynamic range to convert all photons generated during a single muon detection event. However, as previously mentioned, a dark count rate of around 100 kHz is observed at high gain levels of 10^6 (see Table II), which must be considered.

TABLE I. Overview of most important properties of the used organic scintillation material (AdvTECH UK Limited).

Parameter	Value
Light output relative	50%–60% of anthracene
Peak emission	425 nm
Light attenuation	2 000 mm
Refraction index	1.58
Density	$1.05\,\mathrm{g/cm}^3$
Scintillation decay time	2.4 ns
Volumetric size	$250 \times 250 \times 10\,\mathrm{mm}^3$
Surface size	$135\,000\,\mathrm{mm}^2$
Surface finish	All sides polished

TABLE II. Overview of most important properties of the used silicon photomultiplier device ASD-NUV4S-P-40 (CEFLA S.C.).

Parameter	Value
Active area	$4 \times 4\,\mathrm{mm}^2$
Number of cells	9340 pcs
Cell/Total capacitance	90 fF/0.84 nF
Break down voltage	26 V
Over voltage	2–6 V
Gain	10^6
Peak sensitivity	420 nm
Photon detection efficiency	43%
Dark count rate	$50 - 100\,\mathrm{kHz/mm}^2$
Refractive index epoxy resin	1.5115
Spectral transmission	>99% @400–1000 nm

To transmit the generated optical photons from the scintillation plate in one of the detector configurations, several wavelength-shifting (WLS) fibers Y11(200) from Kuraray with an inner diameter of 1 mm were used. The key parameters of the WLS fibers are summarized in Table III. The absorption peak of the WLS fiber at 430 nm closely matches the emission spectrum of the selected scintillator material. Incorporating WLS fibers augments the effective detection area of the SiPMs, enabling the collection of photons in user-defined configurations. The individual fiber lengths were cut to 250 mm to enable photon collection from various surface positions. While the sensitivity peak of the SiPM device listed in Table II does not perfectly align with the WLS fiber emission spectrum, the converted photons still produce electrical signals with approximately 30% efficiency.

Finally, the LTC6268-10 operational amplifier from analog devices was selected for electrical charge conversion, as it offers a high slew rate of up to $1500\,\mathrm{V}/\mu\mathrm{s}$, which is suitable for generating short pulses in the range of several hundred nanoseconds. Furthermore, this FET-based amplifier features an extremely low input bias current and very low input capacitance. Such amplifiers must be carefully designed to ensure stable operation, particularly when the input capacitance varies. The amplifier circuit was implemented using an electronics board developed in-house, which is equipped with multiple input connections for SiPMs. A CAEN

TABLE III. Overview of most important properties of the used wavelength shifting (WLS) fiber Y11(200) (Kuraray).

Parameter	Value
Fiber diameter	1 mm
Cladding modality	Multi-cladding
Emission spectra peak	476 nm
Absorption spectra peak	430 nm
Attenuation length	>3500 mm

21 October 2025 08:11:13

© Author(s) 2025

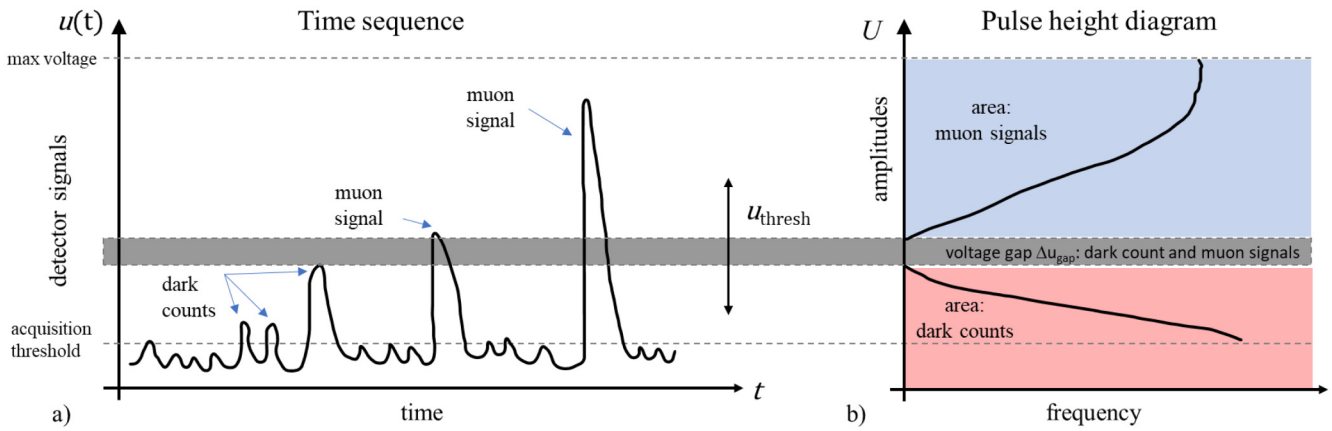


FIG. 1. Functional principle to enable discrimination of muon event and dark count signals using a simple threshold value: (a) time sequence and (b) pulse height diagram.

VX2740 system was employed for data acquisition in combination with WaveDump 2.0 data acquisition software. This system provides a sampling frequency of 125 MHz and a voltage resolution of 16 bit, supporting signal input ranges of ± 2048 mV.

B. Methods

An efficient muon trigger detector must be capable of generating a clearly identifiable and distinguishable signal in response to a muon hit event. It must also effectively suppress or discriminate dark count signals, which are self-triggered in SiPMs due to thermal electrons or overvoltage conditions. Since dark counts are predominantly caused by the activation of up to 10 individual cells, a simple voltage threshold u_{thresh} can be applied to distinguish them from true muon events. Figure 1(a) illustrates the functional principle. The underlying assumption is that a valid muon signal requires the detection of more than 10 optical photons by the SiPM. As shown in Fig. 1(b), the optimal case is a voltage gap Δu_{gap} in the SiPM pulse height spectrum in which neither dark counts nor muon events appear. In such cases, the threshold can easily be defined. However, if no voltage gap is present, the optimal threshold must be chosen as a trade-off between achieving a low dark pulse rate and maintaining sufficient muon detection efficiency.

Due to the isotropic emission of scintillation photons, it is unlikely that the majority will directly strike the surface of the coupled SiPMs. Furthermore, the likelihood of generated optical photons reaching the SiPM surface is reduced further by imperfect reflection conditions at the surfaces of the scintillation plate. This results in a significantly reduced detected photon yield and lower trigger efficiency. To compensate for this loss and improve the reliability of muon event detection, reflective surfaces or coatings are applied to the scintillator surfaces to enhance photon retention and guide more photons toward the SiPM. To investigate the effect of reflective materials on internal photon reflections, three reflector types were evaluated. These included (i) EJ-510 white reflector paint with TiO_2 suspension from Eljen Tech®, (ii) commercially available aluminum foil (shiny side), and (iii) super high-gloss resin-coated photo paper (300 gsm) from Print4Life®. Figure 2 shows the plates covered with the reflective materials they are intended for.

In the case of the reflective paint configuration, the scintillator plate was fully coated with three successive layers, with each layer being applied once the previous one had dried. A typical three-layer coating results in a total thickness of around 0.11 mm. Intentionally leaving space uncovered at the SiPM coupling points allowed for the application of optical grease between the scintillator plate and the SiPM surface, ensuring efficient optical transmission. The same procedure was followed for the aluminum foil and glossy paper configurations. For the aluminum foil configuration, the material was carefully wrapped around the plate and small amounts of optical grease were applied selectively to ensure proper contact between the scintillator and reflective layer. For the glossy paper, sheets were simply cut to size and placed on each surface without using optical grease. While the glossy paper was easier to handle and adhere, the aluminium foil was more challenging to work with, prone to wrinkling and susceptible to tearing during application. The prepared scintillator plates were mounted inside

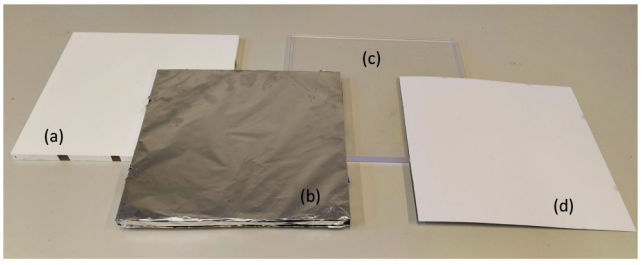


FIG. 2. Scintillation plate covered with (a) reflector paint, (b) aluminum foil, (c) no reflective material, and (d) high-gloss paper.

21 October 2025 08:11:13

© Author(s) 2025

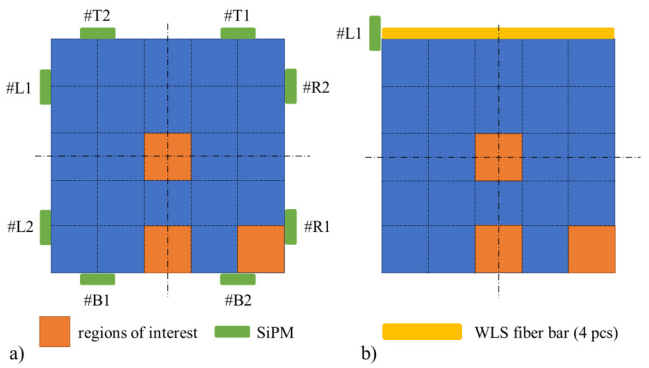


FIG. 3. Sketches of trigger detector configurations to be investigated using (a) various number of SiPM devices and (b) a single SiPM in combination with a WLS fiber clamp.

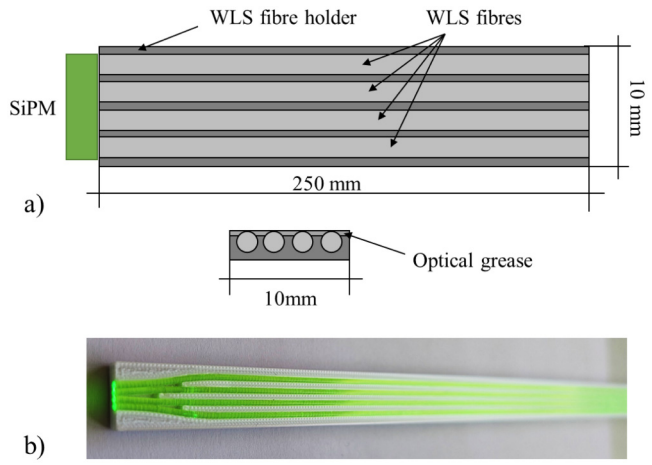


FIG. 4. (a) Sketch and (b) photograph of the WLS fiber clamp for detector *Config 4*.

3D-printed black ABS enclosures. These enclosures featured dedicated notches to accommodate the SiPM surface protrusion and provide space for integrated signal processing electronics.

As shown in Fig. 3 and summarized in Table IV, four SiPM configurations were constructed and evaluated. T, B, L, and R represent the positions of the SiPMs on the top, bottom, left, and right sides of the scintillator plate, respectively. The numbers following the letters indicate the sequential numbering of the SiPMs. In *Config 1*, eight SiPMs are equidistantly mounted on all four sides of the plate. *Config 2* includes four SiPMs, with one mounted on each side face. *Config 3* comprises two SiPMs, both of which are mounted on the same side. The aim of this study is to determine the minimum number of SiPM devices required to achieve a measurable voltage gap Δu_{gap} for reliable muon event discrimination, depending on the reflective surface used.

Finally, *Config 4* is proposed as a general approach for implementing SiPMs in large-area muon scintillation trigger detectors. In this configuration, a single SiPM device is augmented with a wavelength-shifting (WLS) fiber clamp. Details of the WLS fiber clamp assembly are presented in Fig. 4. A 3D-printed or milled bar clamp is used to hold several WLS fibers. This clamp increases the effective active area for scintillation photon collection and can be adapted to various lengths based on application requirements. In *Config 4*, four fibers (Ø1.0 mm) are inserted and secured in a 250 mm-long bar featuring channels that are approximately 1.0 mm wide. These channels are designed to hold the fibers securely within the 3D-printed black enclosure and to guide the collected optical photons toward the active surface of the SiPM [see Fig. 4(b)]. The side facing away from the SiPM has been covered with a reflective material. To ensure efficient photon transfer and minimize losses due to refractive index mismatch, the gap between the WLS fiber clamp and the side face of the scintillator plate is liberally filled with optical grease. This configuration enables the acquisition of scintillation photons generated by muon events over an area of 250 × 250 mm^2 using a single SiPM. If this configuration provides a sufficient voltage gap Δu_{gap}, it could be used as the preferred design for industrial muon trigger detectors, offering robustness and flexible geometry. The design may also be adapted to support WLS fibers of different diameters. For example, SiPM models can be selected whose active areas are optimally matched to the cross-sectional dimensions of the fiber ends.

Configurations in which the SiPM and the WLS fiber were positioned on the top and bottom faces were not considered due to the internal reflection geometry of the scintillation plate. Because the area on the narrow side of the plate is relatively low, a significant proportion of the scintillation light is transported by total reflection on the top and bottom faces and is then collected toward the narrow side faces. Consequently, the detected photon yield is higher on the narrow side faces than on the top and bottom faces.

As the number of SiPM devices connected in parallel to the operational amplifier's input varies, two key quantities are affected: (a) the total input capacitance (see Table II) and (b) the mean probability of a photon interacting with the SiPM surface. Consequently, the feedback loop components (specifically, the capacitance and resistance values) must be adapted for each SiPM configuration listed in Table IV (see Fig. 5). These changes alter both the effective gain and the position of the voltage gap Δu_{gap} between configurations. Consequently, the detector configurations cannot be directly compared with one another. However, it is still

TABLE IV. Overview of detector configurations used for investigation.

Config number	SiPMs count	SiPMs positions	Active area (mm^2)	WLS use
1	8	TBLR1, TBLR2	128	No
2	4	TBLR1	64	No
3	2	T1, T2	32	No
4	1	L1	2500[a]	Yes

[a]The effective collection area includes the area of the WLS fibers in cladding with the sides of the scintillation slab as shown in Fig. 3(b).

21 October 2025 08:11:13

© Author(s) 2025

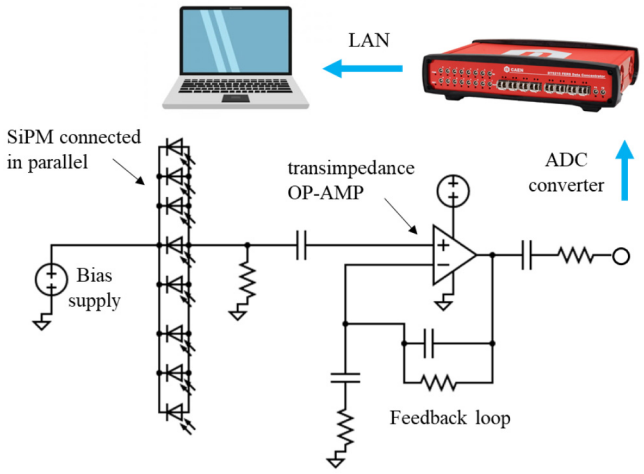


FIG. 5. Sketch of the signal electronics used for readout of numerous SiPM using a single transimpedance amplifier circuit.

feasible to conduct a quantitative evaluation of the effects of different reflective materials for each configuration individually.

C. Experiments

The proposed detector configurations were evaluated using a Muon Coincidence Detector (MCD) system developed in-house. The MCD comprises two identical reference scintillation detectors, each measuring $50 \times 50 \times 10\,\text{mm}^3$ and coated with reflective paint. These reference detectors face each other and are positioned parallel to one another, with the capability to adjust their vertical separation. Each reference detector is read out using a traditional vacuum-glass photomultiplier tube (Hamamatsu). As shown in Fig. 6, when both reference detectors generate a signal simultaneously, this indicates that a muon passed both layers. Such a coincidence event is recorded as a reference muon hit N_{ref} using the CAEN data logger. To ensure comparable muon detection sensitivities, the electronics of both MCD reference detectors were calibrated to match their coincidence pulse height spectra. The CAEN

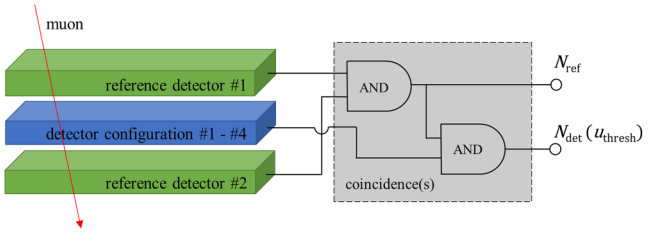


FIG. 6. Principle sketch of coincidence measurements performed to extract the muon detection efficiency of each detector configuration.

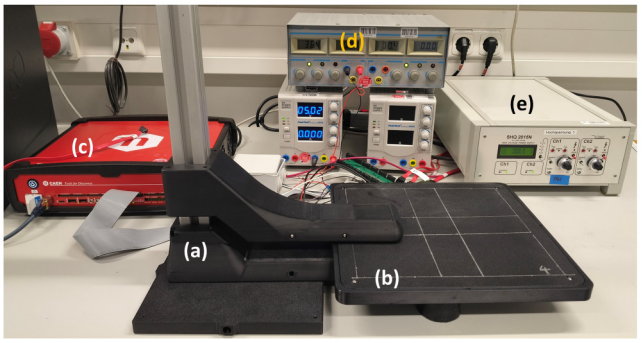


FIG. 7. Photograph of the measurement system for investigations on muon detection efficiency: (a) muon coincidence detector (MCD), (b) scintillation detector configuration to be investigated, (c) CAEN data logger, (d) analog signal voltage supply, and (e) high-voltage supply for photomultiplier tubes.

data logger was set to a trigger threshold of 100 mV to register muon flux events (see Fig. 1). Figure 7(b) shows a photograph of the muon trigger detector positioned between the reference arms of the MCD setup, as depicted in Fig. 7(a).

As shown in Fig. 3, three regions of interest were selected on the scintillation plate for investigation of detection efficiency: the central region, a middle-edge region, and a corner region. The MCD was positioned above each region of interest and coincidences N_{ref} from the MCD were recorded alongside the corresponding signal amplitudes from each scintillation trigger detector configuration, denoted as N_{det} (see Fig. 6). An 8 µs sampling interval was selected to acquire the pulse height spectra for both muon hit events and dark counts. The muon detection efficiency,

$$DE(u_{\text{thresh}}) = \frac{N_{\text{det}}(u_{\text{thresh}})}{N_{\text{ref}}}, \tag{1}$$

can be evaluated from the coincidence pulse height spectra for different threshold values u_{thresh}. Where no voltage gap Δu_{gap} is present, the dark count range overlaps with the muon signal range. Under these conditions, selecting an appropriate u_{thresh} value not only helps to suppress dark counts but also reduces muon detection efficiency.

III. RESULTS AND DISCUSSIONS

The results are structured as follows: First, the start-up procedure for preparing all trigger detector configurations is described. This is followed by an overview of the dark count ranges associated with each detector configuration. Next, the measured detection efficiencies for *Config 1* to *Config 3* trigger detector are presented. Then, the results for *Config 4*, which is assembled exclusively using high-gloss paper, are discussed. Finally, the performance of the WLS fiber clamp-based trigger detector is evaluated. To achieve this, *Config 1* is repurposed as another MCD reference system in order to determine the detection efficiency across the full $250 \times 250\,\text{mm}^2$ scintillation area.

21 October 2025 08:11:13

© Author(s) 2025

A. System assembly and configuration

The trigger detector configuration involves placing a bare, polished scintillation plate into a black, 3D-printed holder. The required number of SiPM devices are optically coupled to the plate's surface using optical grease. The SiPMs are then connected to a transimpedance amplifier via a printed circuit board (PCB), and the resulting signal is transmitted to the CAEN data logger via coaxial cables. The MCD system is positioned in the center of the trigger detector. Its coincidence output is used as a trigger input for the oscilloscope, which is used to visualize signals assigned to muons. Figure 8 shows superimposed voltage signals from an optimized amplifier feedback loop, which ensures that most of the signal amplitudes remain within the amplifier circuit's dynamic range. The measurements are then repeated with the reference muon count fixed at $N_{\text{ref}} = 500$ for a time interval of approximately 25 min, in order to collect a sufficient number of muon hit events.

B. Quantifying dark count rates in detector configurations

Config 1 to *Config 3* were analyzed using a total of 6000 muon hit events, while *Config 4* was examined using 1500 events, each with an 8 μs sampling interval. As expected, differences in both the position and spread of the dark count ranges were observed, primarily due to the varying transimpedance amplifier stages used in each configuration. Nevertheless, the results, shown in Fig. 9, suggest that configurations with larger SiPM active areas produce dark count pulses with lower amplitudes. This demonstrates that using multiple SiPM devices in parallel increases the effective active area, leading to an improved signal-to-noise ratio. Based on the spectra, an initial discriminator threshold value of $u_{\text{thresh}} = 400$ mV is recommended for all trigger detector configurations, effectively excluding the majority of dark counts from muon signal measurements.

It should be noted that *Config 4* was investigated using high-gloss paper as the only reflector, which resulted in a fourfold reduction in the number of recorded events compared to *Config 1* to *Config 3*. Nevertheless, the available data are sufficient to validate the proposed threshold value. It is also worth noting that fewer signals of a comparable high amplitude are registered, which can be attributed to naturally occurring radioactivity.

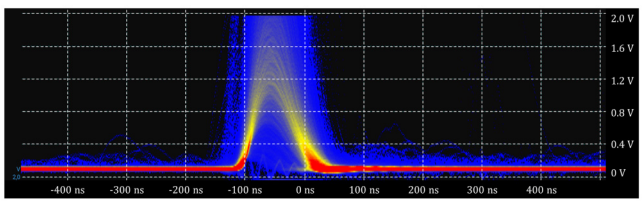


FIG. 8. Oscilloscope screenshot of 50 superimposed muon signals obtained from a trigger detector using coincidence signals of the muon coincidence detector as trigger input.

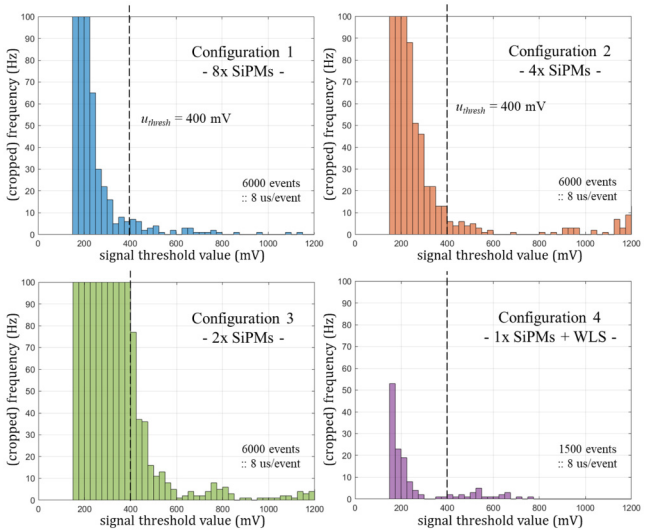


FIG. 9. Dark count spectra of all trigger detector configurations.

C. Investigations with various reflector layers and SiPM configurations

The position and width of the coincidence event pulse height spectra were extracted from the measurements, and the results are shown in Fig. 10. Initially, the signal trends were analyzed in relation to the number of SiPMs used, with a focus on the *Config 1* to *Config 3* trigger detector. For improved visualization, the final histogram bar representing coincidence amplitudes exceeding 2000 mV was cropped from all spectra, as these values fall outside the amplifier circuit's dynamic range. Figure 10 shows the three MCD measurement regions in the rows and the different trigger detector configurations in the columns. The results obtained for the four different reflective materials are displayed in each cell of the figure.

It is evident that configurations without a reflective layer consistently produce the lowest detected photon yield at the SiPMs. Notably, the signal-to-noise ratios are highest and nearly identical for both the aluminium foil and high-gloss paper configurations. Furthermore, the lowest detected scintillation photon yield distributions occur for all trigger detector configurations when the MCD system is positioned at the corner of the scintillation plate. Nevertheless, *Config 3*, which uses only two SiPMs and a bare scintillation plate, still produces signal amplitudes that exceed the dark count threshold value of $u_{\text{thresh}} = 400$ mV.

For the quantitative evaluation, the detection efficiency $DE(u_{\text{thresh}})$ was calculated from the pulse height data using Eq. (1) with threshold values in the range of 200 mV $< u_{\text{thresh}} <$ 1800 mV. As shown in Fig. 11, all configurations except those without a reflective layer achieve a detection efficiency of 80% or higher at $u_{\text{thresh}} = 400$ mV. This indicates that, when combined with a reflective layer, *Config 3* is sufficient for the given scintillation plate geometry to reliably distinguish between dark counts and muon

21 October 2025 08:11:13

© Author(s) 2025

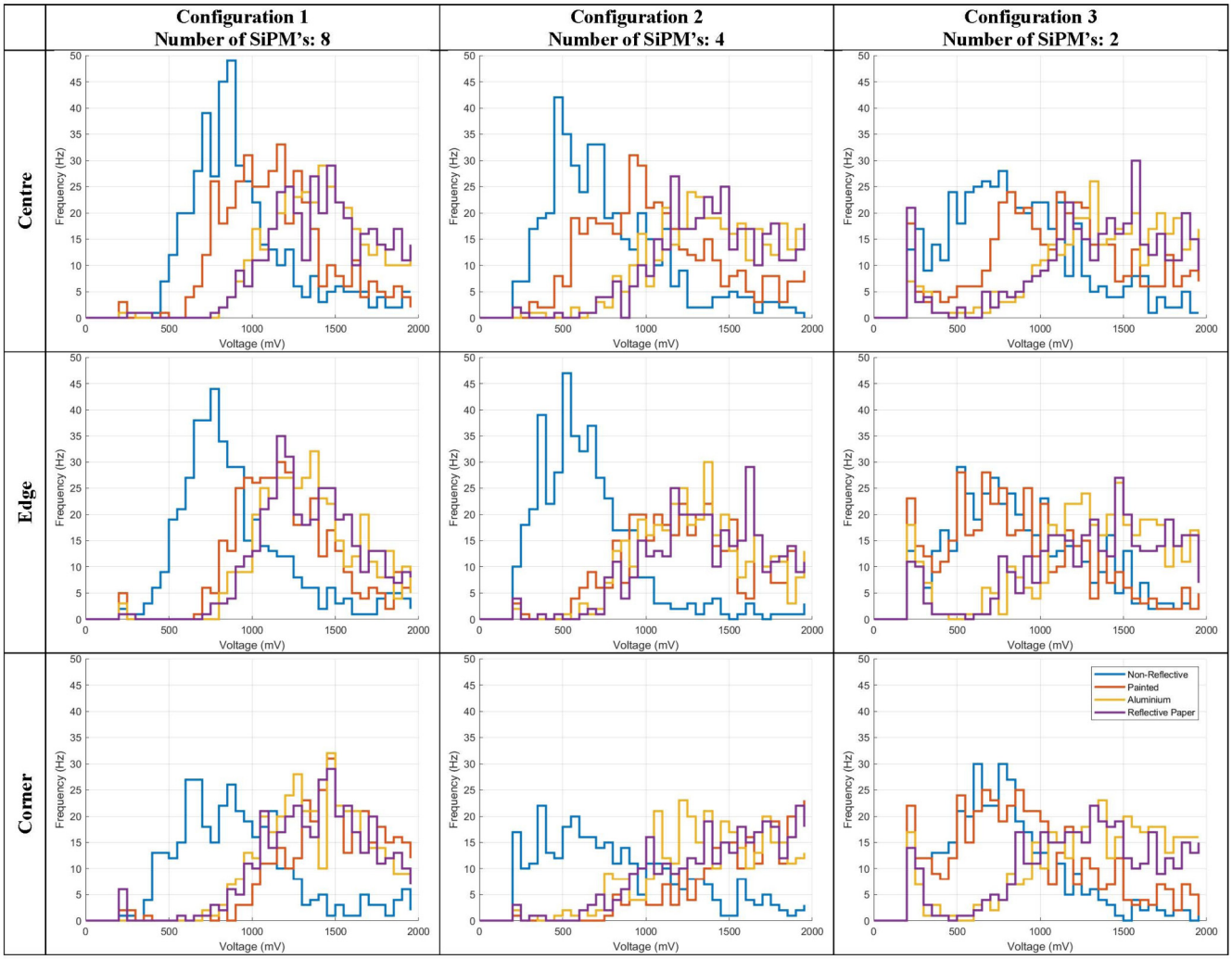


FIG. 10. Comparisons of detected photon yield distribution for trigger detector *Config 1 to 3* at center, edge, and corner measurement locations using the MCD systems.

events while maintaining high detection efficiency. Furthermore, using high-gloss paper for the reflective layer is particularly advantageous as it requires the least effort during assembly.

D. Comparative assessment of detector design with augmented WLS detection area

Next, the detected scintillation photon yield distribution for the *Config 4* trigger detector was investigated. High-gloss paper was applied as the reflective material, and the MCD system was used with $N_{\text{det}} = 500$ coincidence events. The resulting pulse height spectra were then compared with those from the *Config 3* trigger detector. As can be seen in Fig. 12, *Config 4* exhibits a significantly higher detected scintillation photon yield distribution at all measurement positions than *Config 3*. This clearly demonstrates the effectiveness of augmenting the SiPM's active area using a WLS fiber clamp (see Table IV and Fig. 3). Furthermore, the detector efficiency evaluation presented in Fig. 13 shows a detection efficiency of 90% at $u_{\text{thresh}} = 400\,\text{mV}$.

E. Characterization of detector design with augmented WLS detection area

Measurements were conducted to determine (i) the overall detection efficiency of the *Config 4* trigger detector across its entire $250 \times 250\,\text{mm}^2$ active area and (ii) to establish a moderate data rate by selecting an appropriate data acquisition threshold, u_{thresh}.

In the latter case, the WLS-fiber clamp detector was only measured for 10 min using a constant acquisition threshold of $u_{\text{thresh}} = 300\,\text{mV}$, triggering a total of 163 481 events. This is much

21 October 2025 08:11:13

© Author(s) 2025

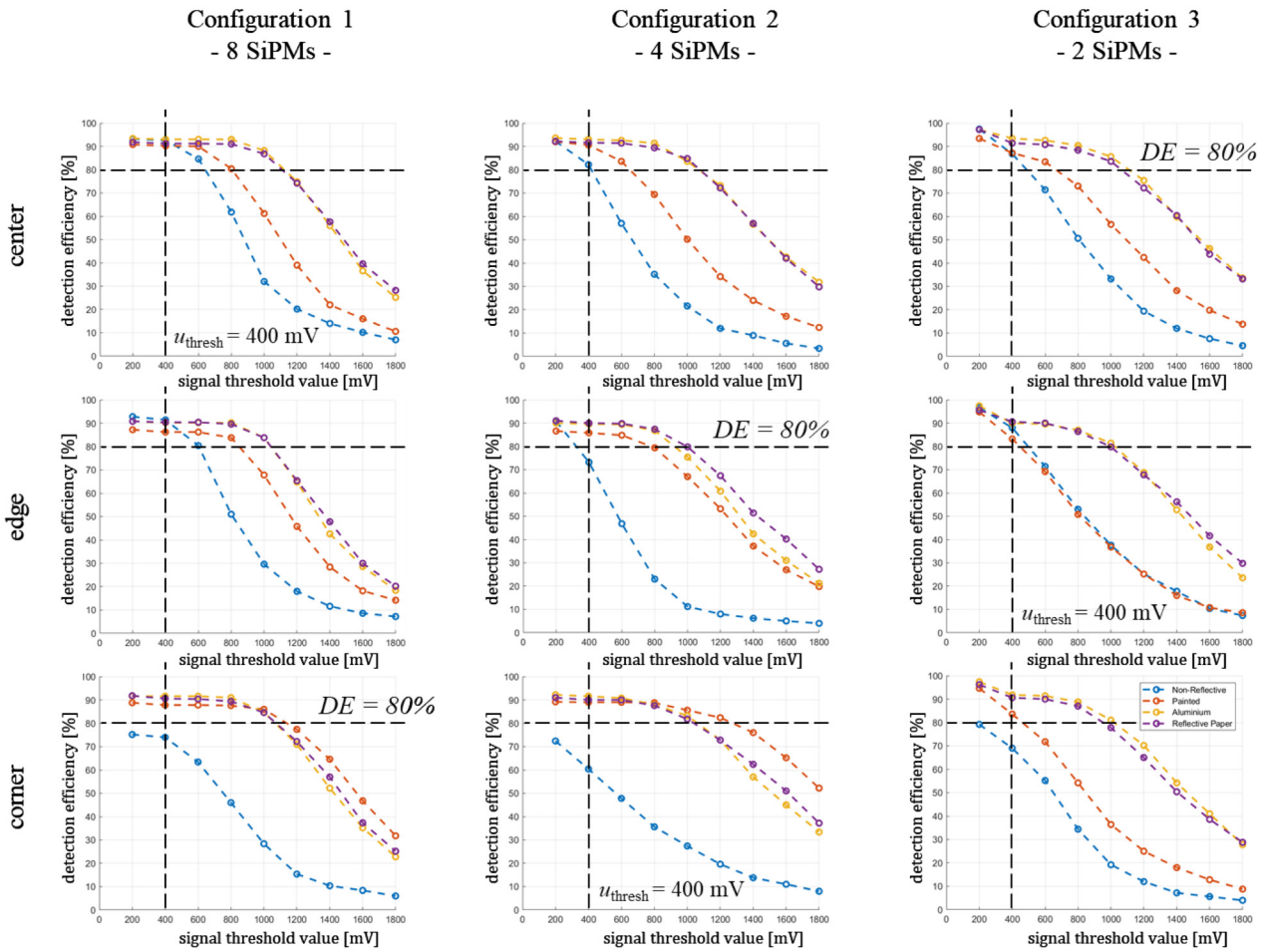


FIG. 11. Comparisons of detection efficiency (DE) for investigated trigger detector configurations 1 to 3 (legend: blue = no reflective layer, orange = reflective paint, yellow = aluminium foil, purple = high-gloss paper).

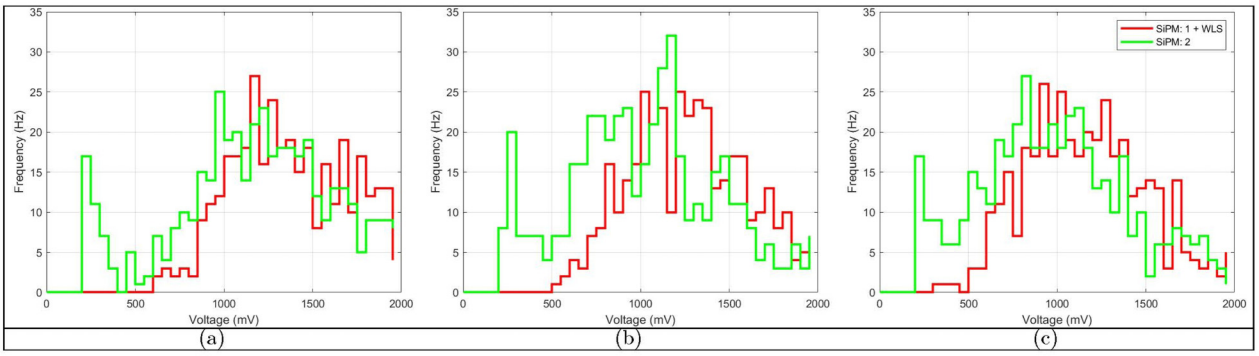


FIG. 12. Pulse height diagrams of the detected scintillation photon yield distribution between trigger detector *Config 3* (green) and *Config 4* (red) obtained at (a) center, (b) edge, and (c) corner locations and using glossy paper as a reflective material.

21 October 2025 08:11:13

© Author(s) 2025

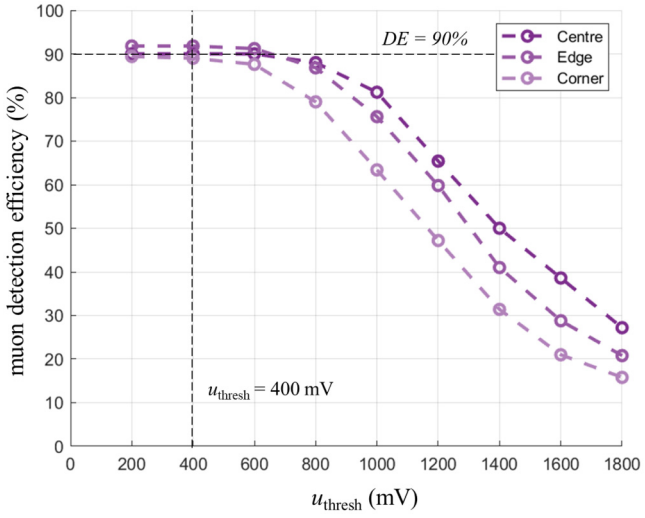


FIG. 13. Detection efficiency of the *Config 4* trigger detector obtained at center, edge, and corner locations and using glossy paper as a reflective material.

higher than the expected number of 6250 muon events at sea level, indicated by the red dashed line in Fig. 14. Incrementally increasing u_{thresh} to 1800 mV in a data post-processing step allows the corresponding number of triggered events to be extracted. As can be seen in Fig. 14, a sharp decrease is observed for $u_{thresh} \leq 500$ mV, while beyond this, the data rate stabilizes and shows only a slight decrease, remaining consistently above the expected muon flux at sea level.

Then, two identical reference trigger detectors, assembled using *Config 1* and covered with high-gloss paper, were exactly

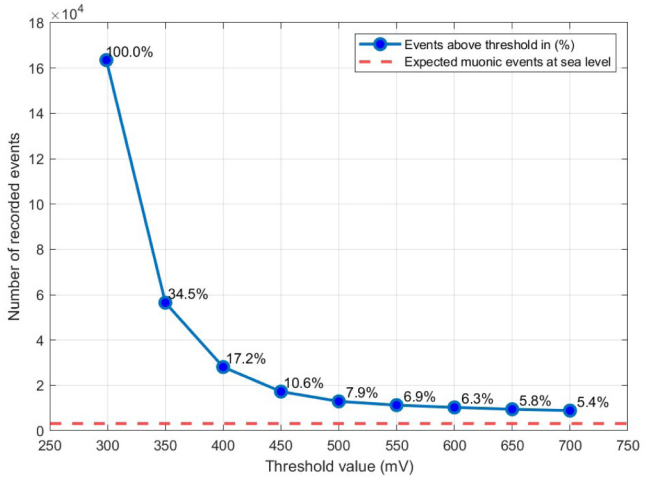


FIG. 14. Investigations on the WLS-fiber detector (*Config 4*) to find an acquisition threshold u_{thresh} that provides moderate data rate.

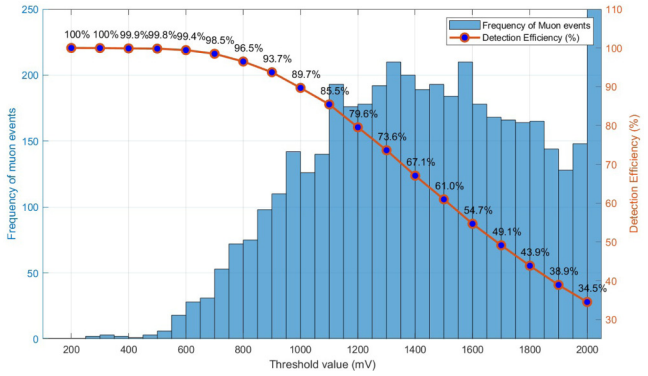


FIG. 15. Merged graph of the coincidence pulse height spectrum of the WLS-fiber detector (*Config 4*) and its calculated detection efficiency *DE*. The data were obtained using two identical reference detectors (*Config 1*) for coincidence triggering.

positioned above and below the WLS-fiber detector. Samples of 8 μs length from all three detectors were recorded when both reference detectors were triggered simultaneously. The measurement duration was 30 min, and the data acquisition threshold for the reference detectors was set to $u_{thresh} = 200$ mV. A total of 6652 muon events were recorded. Figure 15 shows the coincidence pulse height spectra of the WLS-fiber detector and its calculated muon detection efficiency *DE* as a function of u_{thresh}. The graph clearly shows that at a threshold value of $u_{thresh} = 600$ mV, the detection efficiency exceeds 99%.

Figure 14 clearly illustrates the importance of selecting an appropriate threshold value u_{thresh} for data acquisition. When $u_{thresh} < 400$ mV, the number of triggered events increases significantly due to dark counts, which do not contribute additional muon signals. Conversely, for $u_{thresh} > 400$ mV, the majority of dark counts are effectively excluded. However, beyond a certain threshold, the muon detection efficiency begins to decline. Based on these observations, a valid voltage range between 400 and 1200 mV is identified for the WLS-based trigger detector configuration. Within this range, a muon detection efficiency of 80% or higher can be achieved while successfully suppressing most of the dark count signals.

IV. CONCLUSION

This study systematically evaluated the muon detection performance of four muon trigger detector configurations using an organic scintillation plate measuring $250 \times 250 \times 10\,\text{mm}^3$ and silicon photomultiplier (SiPM) devices with an active area of $4 \times 4\,\text{mm}^2$. Various reflective materials, including optical reflector paint, aluminum foil, and high-gloss paper, were investigated to enhance the yield of the detected scintillation photons and improve the overall quality of the signal. Of these materials, high-gloss paper produced the most favorable results for our detector geometry, offering excellent reflectivity and ease of integration for scalable assembly.

21 October 2025 08:11:13

© Author(s) 2025

Detector configurations comprising 8, 4, and 2 SiPMs were systematically evaluated. This demonstrated that muon-induced signals remain sufficiently strong to be distinguished from dark counts even with only two SiPMs, ensuring reliable detection performance. Based on these findings, a compact, cost-effective detector design using a single SiPM in combination with wavelength-shifting (WLS) fibers embedded in bar clamps assembled in-house was introduced. This configuration reduced sensor requirements and achieved the highest signal-to-noise ratio of all tested designs, as well as a position-independent detection efficiency of up to 90%. An acquisition threshold voltage range between 400 and 1200 mV was identified as optimal for effectively suppressing thermally induced dark counts while preserving muon detection efficiencies of over 80%. Using high-gloss paper as a reflective material further improved light collection efficiency and offered practical advantages in terms of assembly and scalability.

Overall, the WLS-augmented configuration offers a robust, modular, and flexible solution for designing large-area muon trigger detectors that can be adapted to complex geometries and industrial-scale applications. Together, the proposed evaluation methodology and detector architecture offer a reliable foundation for the future development of high-efficiency muon detection systems that can be tailored to diverse operational environments.

ACKNOWLEDGMENTS

We thank our colleagues: Martin Löschau for useful discussions, hardware developments, mechanical design, and construction of the detector parts; and Ronny Berger for useful discussions and for performing preliminary tests.

FUNDING

This work was funded by the Helmholtz Association (Germany) under Grant No. KA-TVP-37 and by the Federal Ministry for the Environment, Nature Conservation, Nuclear Safety and Consumer Protection (BMUV) under Grant Nos. 1501661 and 1501606A.

AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

All authors have read and agreed to the published version of the manuscript.

G. Rao: Conceptualization (lead); Data curation (lead); Formal analysis (equal); Investigation (lead); Methodology (equal); Software (lead); Validation (equal); Writing – original draft (lead); Writing – review & editing (equal). **A. Bieberle:** Conceptualization (equal); Formal analysis (lead); Methodology (supporting); Software (supporting); Visualization (lead); Writing – review & editing (equal). **U. Hampel:** Funding acquisition (lead); Project administration (lead); Resources (lead); Writing – review & editing (equal). **M. Wagner:** Conceptualization (equal); Methodology (supporting); Software (equal); Supervision (lead); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

REFERENCES

[1]ATLAS Collaboration, "The ATLAS experiment at the CERN large hadron collider," J. Instrum. **3**, S08003 (2008).
[2]A. Aab, P. Abreu, M. Aglietta *et al.*, "Prototype muon detectors for the AMIGA component of the pierre auger observatory," J. Instrum. **11**, P02012 (2016).
[3]A. Castellina and The LHAASO Collaboration, "Augerprime: The pierre auger observatory upgrade," EPJ Web Conf. **210**, 06002 (2019).
[4]PANDA Collaboration, "Technical progress report for the PANDA experiment at FAIR," Tech. Rep. (FAIR, 2009). doi:10.1140/epja/i2008-10635-8.
[5]ALICE Collaboration, "The ALICE experiment at the CERN LHC," J. Instrum. **3**, S08002 (2008).
[6]L. W. Alvarez, J. A. Anderson, F. E. Bedwei *et al.*, "Search for hidden chambers in the pyramids by cosmic-ray absorption," Science **167**, 832–839 (1970).
[7]R. Alfaro, E. Belmont-Moreno, A. Cervantes, V. Grabski, J. M. López-Robles, L. Manzanilla, A. Martínez-Dávalos, M. Moreno, and A. Menchaca-Rocha, "A muon detector to be installed at the Pyramid of the Sun," Rev. Mex. Fís. **49**, 54–59 (2003).
[8]K. Morishima, M. Kuno, A. Nishio *et al.*, "Discovery of a big void in Khufu's pyramid by observation of cosmic-ray muons," Nature **552**, 386–390 (2017).
[9]H. Miyadera, J. Borozdin, E. Guardincerri *et al.*, "Imaging fukushima daiichi reactors with cosmic-ray muons," AIP Adv. **3**, 052133 (2013).
[10]H. K. M. Tanaka, M. Nishiyama, S. Kusagaya, and H. Shinohara, "Cosmic-ray muon imaging of magma in a conduit: Degassing process of Satsuma-Iwojima Volcano," Geophys. Res. Lett. **36**, L01304, https://doi.org/10.1029/2008GL036451 (2009).
[11]J. Wu, W. Huang, R. Cao, Q. Chang, W. Ding, J. Huang, L. Li, X. Li, H. Mei, C. Mo, *et al.* "Muonslab: A plastic scintillator based detector for muon measurement in the deep ocean," arXiv:2501.17639 (2025).
[12]J. M. Durham, D. Poulson, J. Bacon *et al.*, "Verification of spent nuclear fuel in sealed dry storage casks via measurements of cosmic-ray muon scattering," Phys. Rev. Appl. **9**, 044013 (2018).
[13]T. Braunroth, N. Berner, F. Rowold, M. Péridis, and M. Stuke, "Muon radiography to visualise individual fuel rods in sealed casks," EPJ Nucl. Sci. Technol. **7**, 12 (2021).
[14]D. Poulson, B. Morris, F. Garcia *et al.*, "Tomographic imaging with cosmic-ray muons: A review," Rep. Prog. Phys. **80**, 096902 (2017).
[15]M. Niedermeier, J. Müller, L. Roth, T. Schmitt, and P. G. Trosseille, "Development of a modular muon tomography system for industrial applications," J. Instrum. **19**, P02024 (2024).
[16]S. Barnes, A. Georgadze, A. Giammanco *et al.*, "Cosmic-ray tomography for border security," Instruments **7**, 13 (2023).
[17]R. L. Workman, V. D. Burkert, V. Crede *et al.*, "Review of particle physics 2022," Prog. Theor. Exp. Phys. **2022**, 083C01.
[18]D. Autran, S. Marez, J. King *et al.*, "Status of the muon imaging system of the los alamos neutron science center proton radiography facility," IEEE Trans. Nucl. Sci. **65**, 1654–1660 (2018).
[19]Glenn F. Knoll, *Radiation Detection and Measurement*, 4th ed. (John Wiley & Sons, 2010).
[20]M. Green, *Particle Detectors: Fundamentals and Applications* (Springer, 2008).
[21]L. Bonechi, R. D'Alessandro, and A. Giammanco, "Atmospheric muons as an imaging tool," Rev. Phys. **5**, 100038 (2020).
[22]N. Panchal, S. Mohanraj, A. Kumar *et al.*, "A compact cosmic muon veto detector and possible use with the iron calorimeter detector for neutrinos," J. Instrum. **12**, T11002 (2017).

21 October 2025 08:11:13

© Author(s) 2025

[23]M. Saraf, P. R. Chinnappan, A. Deodhar, M. Jangra, J. Krishnamoorthi, G. Majumder, V. Padmavathy, K. C. Ravindran, R. Shah, R. Shinde, and B. Satyanarayana, "Design, fabrication and large-scale qualification of cosmic muon veto scintillator detectors," J. Instrum. **18**, P05003 (2023).
[24]R. Shah, G. Majumder, and P. Shukla, and INO Collaboration, "Expected performance of cosmic muon veto detector," J. Instrum. **19**, P07025 (2024).
[25]W. J. Jo, S. J. An, H. I. Kim *et al.*, "Design and characterization of a portable cosmic-ray muon detector using silicon photomultipliers," J. Korean Phys. Soc. **66**, 585–591 (2015).
[26]C. Park, M. K. Baek, I. Kang *et al.*, "Design and characterization of a muon tomography system for spent nuclear fuel monitoring," Nucl. Eng. Technol. **54**, 601–607 (2022).
[27]C. Park, K. B. Kim, M. K. Baek *et al.*, "Performance of an upgraded cosmic muon tomography system," Nucl. Eng. Technol. **55**, 1009–1014 (2023).
[28]D. G. Michael, P. Adamson, T. Alexopoulos *et al.*, "The magnetized steel and scintillator calorimeters of the MINOS experiment," Nucl. Instrum. Methods Phys. Res. Sect. A **596**, 190–228 (2008).
[29]T. P. Reinhardt, S. Gohl, S. Reinicke *et al.*, "Silicon photomultiplier readout of a monolithic $270 \times 5 \times 5\,\mathrm{cm}^3$ plastic scintillator bar for time of flight applications," Nucl. Instrum. Methods Phys. Res. Sect. A **816**, 16–24 (2016).
[30]F. Arneodo, M. L. Benabderrahmane, S. Dahal, A. Di Giovanni, L. Pazos Clemens, A. Candela, M. D'Incecco, D. Sablone, and G. Franchi, "Muon tracking system with silicon photomultipliers," Nucl. Instrum. Methods Phys. Res. Sect. A **799**, 166–171 (2015).
[31]D. E. Groom, N. V. Mokhov, and S. I. Striganov, "Muon stopping power and range tables 10 MeV–100 TeV," At. Data Nucl. Data Tables **78**, 183–356 (2001).

21 October 2025 08:11:13

© Author(s) 2025